Pertemuan 5 – API Gateway Sederhana

1. Tujuan API Gateway dibuat untuk
   a. Menyatukan semua service dalam 1 pintu

      Sebelumnya User menggunakan localhost:8001, Product menggunakan localhost:8002 dan Order menggunakan localhost:8003. Mengakibatkan client harus akses banyak URL, sekarang semua lewat Gateway (misalnya localhost:8000).
   b. Menyederhanakan akses client

      Client cukup akses /users, /products dan /orders, tidak perlu tahu service mana di belakangnya
   c. Sebagai "perantara" (proxy)

      Gateway menerima request dari client , kemudian meneruskan ke service yang sesuai selanjutnya mengembalikan response ke client. Ini disebut Request Forwarding / Reverse Proxy sederhana.

2. Code gateway/main.py

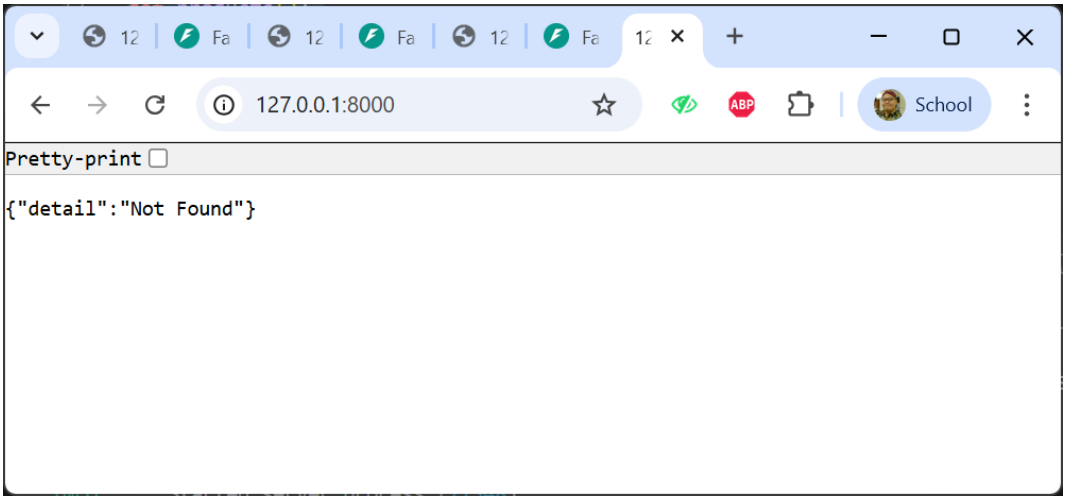
   Silahkan tambahkan folder gateway kemudian baru buat file main.py, sehingga menjadi seperti *gateway/main.py*. Untuk code main.py dapat dilihat seperti dibawah ini.

```
from fastapi import FastAPI
import requests

app = FastAPI()

# Forward request ke user service
@app.get("/users")
def users():
    return requests.get("http://localhost:8001/users").json()

# Forward ke product service
@app.get("/products")
def products():
    return requests.get("http://localhost:8002/products").json()

# Forward ke order service
@app.get("/orders")
def orders():
    return requests.get("http://localhost:8003/orders").json()
```

   Sedangkan untuk menjalankannya dengan perintah sebagai berikut.
   uvicorn gateway.main:app --port 8000 --reload

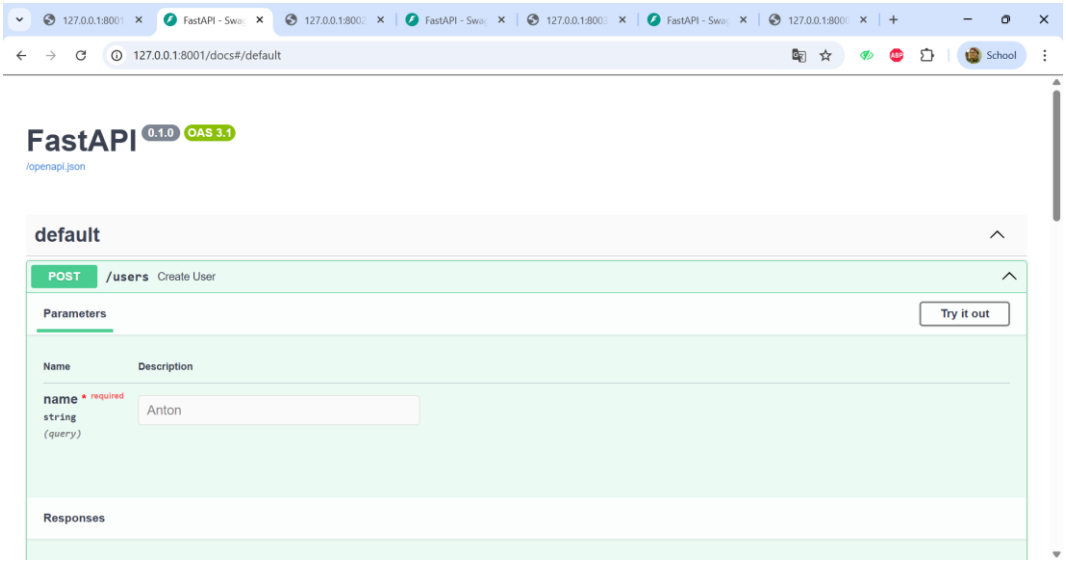
```
PS D:\DosenQ\Mengajar\20252\Sistem Terdistribusi\Praktikum> uvicorn gateway.main:app --port 80
00 --reload
INFO:     Will watch for changes in these directories: ['D:\\DosenQ\\Mengajar\\20252\\Sistem T
erdistribusi\\Praktikum']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [15708] using StatReload
INFO:     Started server process [27348]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
```

untuk hasil waktu diakses dibrowser akan seperti gambar dibawah ini.

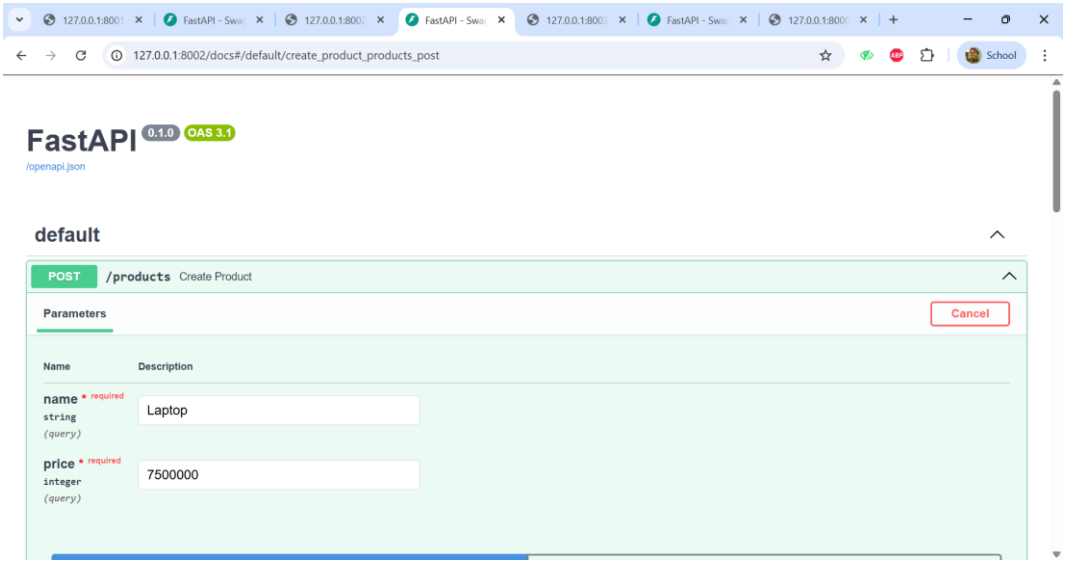


Untuk mengetest gateway http://127.0.0.1:8000/ (akses utama) : Pastikan semua service jalan dan pastikan data sudah di inputkan serta saling berelasi

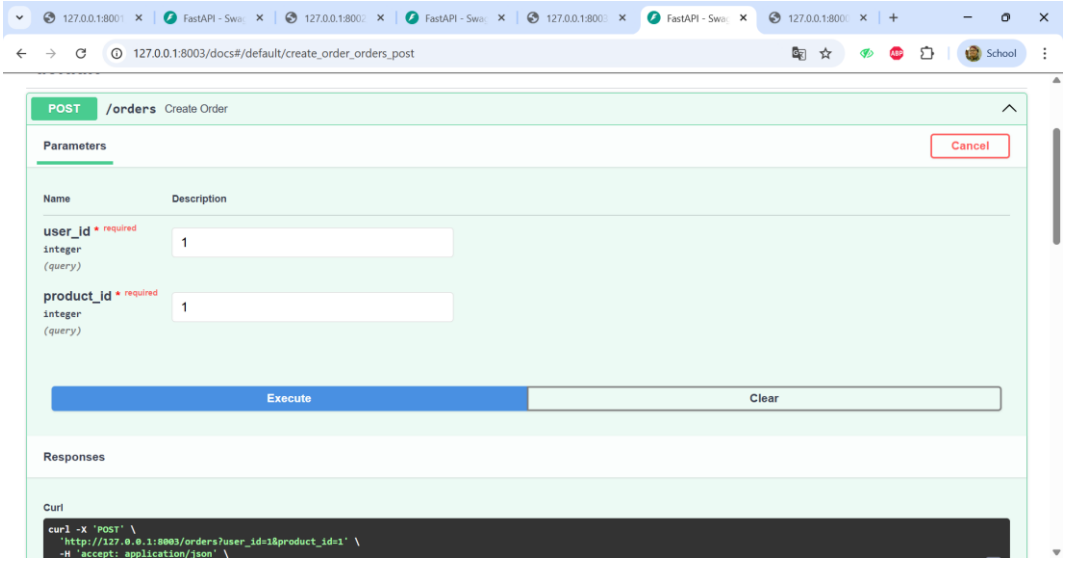
- http://127.0.0.1:8001/users



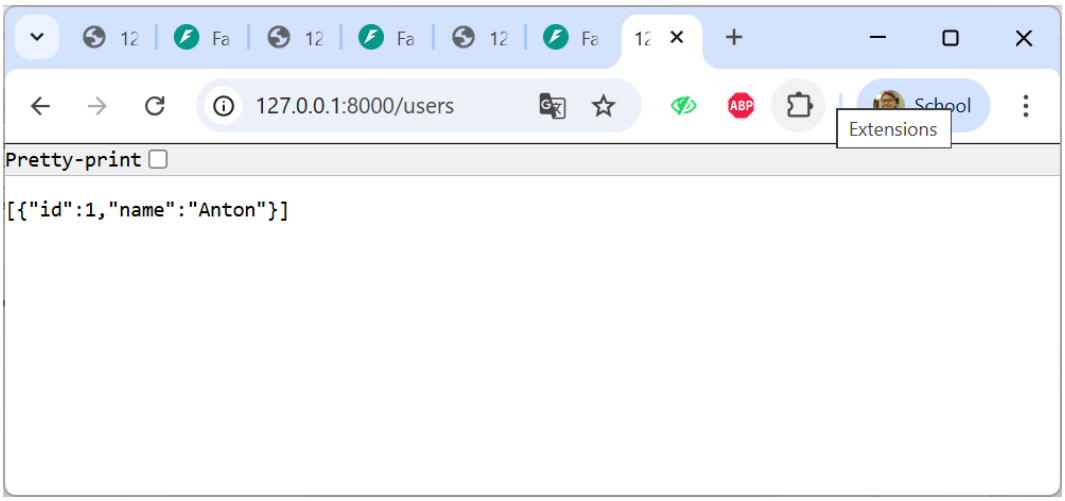
- http://127.0.0.1:8002/products



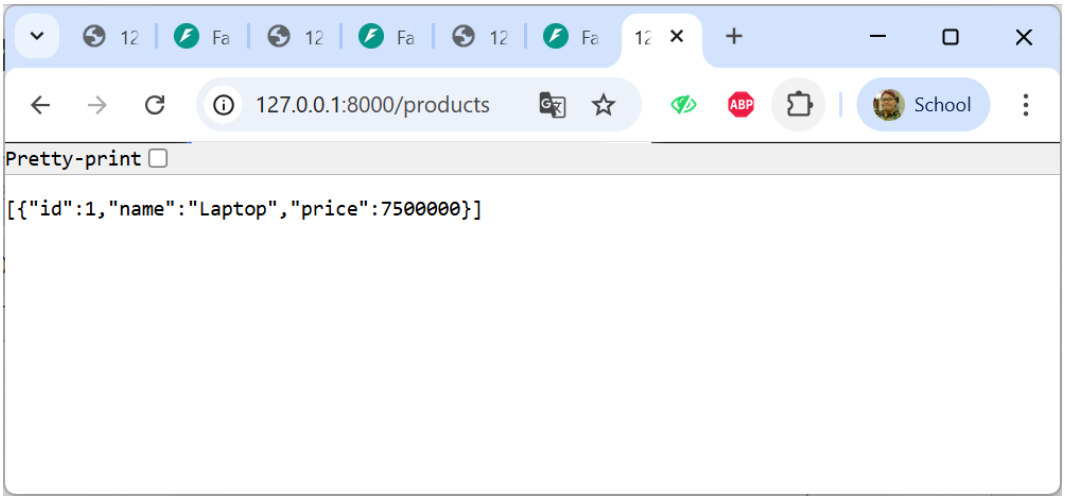
- http://127.0.0.1:8003/orders



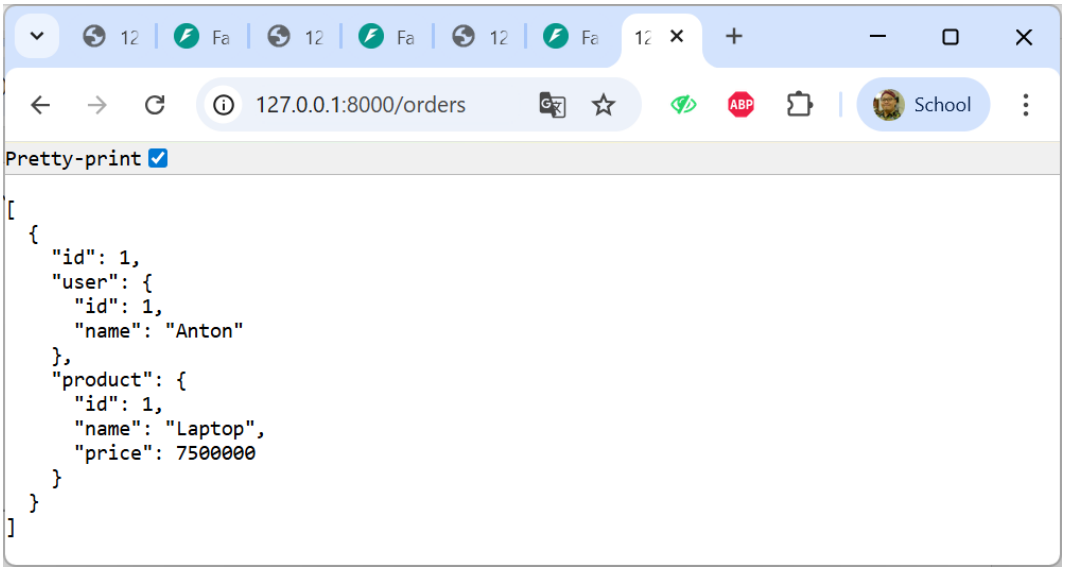
- Selanjutnya test gateway Users



- Gateway Products



- Gateway Orders




```
[
  {
    "id": 1,
    "user": {
      "id": 1,
      "name": "Anton"
    },
    "product": {
      "id": 1,
      "name": "Laptop",
      "price": 7500000
    }
  }
]
```